UNIVERSITY OF ESWATINI

Department of Computer Science

FNB eWallet to MTN MoMo Transfer System

Problem Statement

TEAM MEMBERS

Musawenkhosi Magongo – Project Manager
Setsembele Magagula – Designer
Bandile Lokothwayo – Programmer
Sambulo Mdluli – Tester

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1. Background

The project addresses access to funds received through FNB eWallet in rural areas of Eswatini. It proposes an additional option in the existing Unstructured Supplementary Service Data (USSD) menu. In the rural communities motivating this project, MoMo agents are more readily available nearby than FNB ATMs or CashPlus agents. Customers may therefore have to walk long distances to reach an eWallet cash-out point even when a MoMo agent is close to them.

The proposed service would add a **Send to MoMo account** option within the eWallet menu accessed by dialling ***130#**. The customer would select a MoMo account of their choosing, including their own. Once the transfer is complete, the recipient can use the funds through MoMo or withdraw them at a nearby agent with cash available.

Customer Access Scenario

Stage	Description
Source funds	Customer receives money through FNB eWallet.
Access difficulty	The nearest FNB ATM or CashPlus agent may require a long walk or journey.
Nearby alternative	A MoMo agent is closer to the customer's community.
Proposed route	Transfer from the eWallet USSD menu to a chosen MoMo account, then use or withdraw the funds through MoMo.

Table 1: Rural access scenario motivating the project

Information Required from FNB and MTN

Topic	Information required
Rural access	Distances, travel times and availability of ATMs, CashPlus agents and MoMo agents in selected communities.
Transfer requirements	Verification, recipient validation, limits, charges and withdrawal PIN validity.
USSD integration	Placement of the new menu option, session handling and approved test interfaces.
Customer experience	Current withdrawal journeys and the convenience of a nearby MoMo agent.

Table 2: Information needed to refine and evaluate the proposed service

2. The Problem

For rural customers in Eswatini, receiving money through eWallet does not always mean that the money is conveniently accessible. FNB ATMs and CashPlus agents can be far from their homes, requiring long walks or additional transport costs to withdraw funds. MoMo agents are more accessible in the communities targeted by this project, but the customer needs a way to move the eWallet funds into a chosen MoMo account to use that nearby service.

The central problem is therefore **distance and convenience when accessing money**. The proposed transfer option aims to reduce dependence on travelling to a distant eWallet cash-out point by making the funds available through MoMo.

Issue	Description
Long distances	Rural customers may have to walk far to reach an FNB ATM or CashPlus agent.
Travel time and cost	Collecting money can take time away from daily activities and may require paid transport.
Uneven access	A nearby MoMo agent may be more convenient than the available eWallet cash-out points.
Access to nearby services	Customers need a direct way to move eWallet funds into MoMo to use the closer agent network.
Transfer reliability	The new option must avoid duplicate transfers and give a clear result when processing is interrupted.

Table 3: Access problems addressed by the proposed transfer option

Withdrawal PIN Expiry

A further inconvenience identified in the project brief is the limited validity of an eWallet withdrawal PIN. If it expires before the customer withdraws, obtaining another PIN adds a step to accessing the money. Transferring received funds directly into the customer's own MoMo account would avoid the eWallet cash-withdrawal step and its temporary PIN, subject to the transfer authorisation agreed with FNB.

This concerns the withdrawal PIN, not an assumption that the funds disappear when that PIN expires. The applicable validity period and renewal process in Eswatini will be confirmed with FNB.

Research question: How can a transfer option within the eWallet USSD menu enable rural customers to move funds to a chosen MoMo account reliably and reduce the distance and inconvenience involved in accessing those funds?

The evaluation will document the distances and experiences in selected communities; the project does not assume the same access conditions in every part of Eswatini.

3. Proposed Solution

Add a **Send to MoMo account** option to the eWallet service accessed through ***130#**. The customer completes the transfer through sequential USSD prompts on their phone.

Transferring to One's Own MoMo Account

For example, a customer receives E500 through eWallet and already has a MoMo account. They select **Send to MoMo account**, enter their own MoMo number and the amount, verify the details and confirm. The funds then become available in their MoMo account for payments or withdrawal at a nearby agent, without first withdrawing cash from eWallet.

The academic prototype will simulate this menu journey and the underlying transfer using test accounts and balances. Live menu integration will depend on approval and access from the service providers.

How It Works

STEP 1: Access the eWallet Menu

- The customer dials ***130#** to access the eWallet service.

STEP 2: Select Send to MoMo Account

- The customer selects the additional **Send to MoMo account** option.

STEP 3: Enter the Recipient's Number

- The customer enters the MoMo number of their chosen recipient, which may be their own MoMo account.

STEP 4: Enter the Transfer Amount

- The customer enters the amount to send. The system checks the recipient, amount, transfer limits and balance needed to cover the amount and any fee.

STEP 5: Verify Before Sending

- The customer reviews the recipient number, amount and any applicable fee, then confirms or cancels. Required authorisation follows the provider-approved process.

STEP 6: Send and Display the Result

- The system processes the transfer and displays a sent confirmation only after the MoMo credit is confirmed.
- An uncertain outcome remains pending. A definite failure after debit triggers controlled recovery, with a reversed status only after the source funds are restored.

4. Features by User Role

For Customers

- Dial ***130#** and select **Send to MoMo account** from the eWallet menu.
- Enter the chosen recipient's MoMo number, followed by the transfer amount, through USSD prompts.
- Review the transfer details before confirming.
- Receive a transaction reference and a clear completed, pending, failed or reversed result.
- Send to their own MoMo account for personal use or to another recipient's MoMo account. The recipient can withdraw at a nearby agent where cash is available.

For Support Staff

- Search authorised transaction records by reference.
- Inspect timestamps, status changes and recorded failure reasons.
- Identify pending transfers requiring investigation.
- Record investigation notes without directly editing balances.

For System Administration

- Manage prototype accounts and access permissions.
- Configure simulated service responses and test limits.
- Monitor pending transfers, failures and recovery attempts.
- Export test reports and review the audit trail.

For the Project Test Team

- Run repeatable success, rejection, duplicate and timeout scenarios.
- Compare the expected balances and statuses with actual results.
- Reset isolated test data between demonstrations.

5. Technology Stack

The following is a proposed stack for the academic prototype. Final choices will depend on the agreed requirements and the team's development environment.

Component	Technology	Purpose
USSD simulator	HTML, CSS, JavaScript	Reproduce the customer menu journey in a browser for testing.
Backend	Python with Flask	USSD session handling, authorisation, validation and transfer coordination.
Data storage	MariaDB	Store test accounts, transaction states, balance entries and audit records.
Service simulation	Separate mock HTTP interfaces	Model eWallet debit, MoMo credit and status queries without production access.
Consistency controls	Database transactions and unique request keys	Protect individual updates and prevent repeated requests from moving funds twice.
Testing	pytest and scripted scenarios	Verify balances, state transitions, recovery and access controls.
Version control	Git and GitHub	Track changes and support team collaboration.
Documentation	LaTeX and Markdown	Record requirements, design decisions and evaluation results.

Table 4: Proposed technology stack

The browser-based simulator reproduces the USSD journey for demonstration. A separate staff dashboard supports authorised transaction review. A live service would connect the menu handler to an approved USSD gateway and provider interfaces.

Mock interfaces will model the source debit, destination credit and status checks. The backend will coordinate these operations and recover interrupted transfers. Actual provider interfaces and integration requirements remain subject to agreement with FNB and MTN.

6. Expected Outcomes

Outcome	Benefit
Potential reduction in travel	Demonstrates a route to a nearby MoMo agent and evaluates its potential to reduce travel to an ATM or CashPlus agent.
Working USSD prototype	Demonstrates the journey from dialling *130# to verification and transfer confirmation.
Direct transfer to own MoMo	Lets an eWallet recipient move received funds into their own MoMo account without a cash withdrawal and deposit.
Less reliance on withdrawal PINs	Avoids the eWallet cash-out step and the inconvenience of renewing an expired withdrawal PIN.
Clear transaction status	Helps the user understand whether a transfer is pending, completed, failed or reversed.
Consistent balance records	Demonstrates that success, retries and recovery produce the expected financial state in the simulation.
Traceable transactions	Supports investigation using references, timestamps and state changes.
Access and usability findings	Compares withdrawal travel distances and assesses use of the USSD prompts and status messages.
Documented limitations	Separates demonstrated software behaviour from requirements for a future live integration.

Table 5: Expected project outcomes

The final deliverables will include the prototype, source code, requirements and architecture documentation, repeatable test scenarios, and an evaluation report. Any claims about time saved or customer benefit will be based on collected evidence rather than assumed results.

7. Sample USSD Menu Designs

eWallet Menu and Transfer Prompts

The following screens illustrate the proposed sequence. Menu numbering, exact wording and verification requirements will be agreed with FNB and MTN. Existing eWallet options remain in place.

1. Access the service
Dial *130# to access eWallet.

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2. Select the new menu option
eWallet menu: [existing options]
Send to MoMo account

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3. Enter the recipient's number
Enter recipient's MoMo number:
[Customer enters the chosen number.]

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4. Enter the amount
Enter transfer amount (E):
[Customer enters 500.00.]

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5. Verify before sending
Send E500.00 to [recipient's MoMo number]?
Fee: E0.00 Total debit: E500.00
1. Confirm 2. Cancel

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6. Display the successful result
Sent: E500.00 to [recipient's MoMo number].
Reference: TX-DEMO-0001

Pending and Unsuccessful Transfers

A pending result means that processing is unresolved; "Sent" is shown only after credit is confirmed. A failed result explains why the request was rejected. A reversed result confirms that debited funds were restored.

The values above are illustrative. Provider-required authorisation forms part of verification. Cancelling before authorisation leaves both balances unchanged.

8. Constraints

- **Agent availability:** The convenience of withdrawal depends on the distance to a MoMo agent, operating hours and available cash. These conditions will be recorded in selected rural communities.
- **Prototype scope:** The initial system covers eWallet-to-MoMo transfers only. Reverse transfers are outside the initial scope.
- **No production access:** Use simulated services unless the providers explicitly supply an approved test environment. The project assumes no partnership agreement or production API access.
- **PIN and transfer verification:** Confirm withdrawal PIN validity and transfer authorisation with FNB. The proposed transfer avoids cash withdrawal but still requires the agreed security checks.
- **Test data only:** Use fictitious accounts and balances. Do not request real wallet PINs, withdrawal credentials or customer funds.
- **Connectivity:** The intended customer channel requires mobile network coverage and USSD support, rather than customer internet access. The simulator and backend need their own connection. Session expiry or a dropped connection must not be treated as proof that an authorised transfer failed. A provider-approved status enquiry or notification mechanism will be needed when the session ends before the result is displayed.
- **Development period:** The proposed period is six weeks, subject to agreement on requirements and available resources.
- **Security and access:** Access to transaction records must be restricted by role. Credentials and test configuration must be kept out of public source repositories.
- **Future deployment:** A live service would require separate provider approval and confirmation of operational, security and regulatory requirements.
- **Evaluation limits:** Simulation results cannot establish production performance or prove that a commercial integration is feasible.

9. Success Criteria

The prototype will be evaluated using test accounts and simulated balances:

- The customer can follow the simulated *130# eWallet menu, select Send to MoMo account, enter the recipient number, enter the amount, verify and confirm, and receive the result.
- A customer who receives funds through eWallet can transfer them to their own MoMo number. Tests also cover a different recipient; matching source and destination phone numbers must not be rejected solely because they belong to the same person.
- Cancelling or allowing the session to expire before authorisation leaves both balances unchanged.
- Invalid recipients, non-positive amounts, exceeded limits and insufficient funds (including any configured fee) are rejected before debit.
- For a fee-free E500 transfer from an E1,000 source balance to an E0 destination balance, the final balances are E500 and E500, with one completed transaction.
- Submitting the same request more than once produces no additional debit or credit.
- When credit definitely fails after debit and reversal succeeds, the source returns to E1,000, the destination remains E0 and the status is reversed.
- If the session ends after authorisation or a credit response is lost, the recorded transfer remains traceable. An unknown outcome stays pending until reconciliation establishes one consistent result.
- An unsuccessful reversal remains unresolved and visible for investigation; it is not incorrectly labelled reversed.
- A restart during processing does not erase the transaction or cause a repeated debit when processing resumes.
- The USSD session is linked to the authorised source wallet. Customers cannot access another customer's funds or transfer details; staff access follows assigned permissions.
- Each transfer has a reference, timestamps and an audit trail sufficient to explain its final status.
- A usability exercise records completion of the six-step USSD journey, input errors and understanding of status messages.
- An access comparison records distance and travel time to FNB ATMs or CashPlus agents and nearby MoMo agents in selected rural communities.
- The final report records stakeholder feedback, test results, unresolved issues and the limits of the simulation.

10. Roles and Responsibilities

Role	Name	Responsibilities
Project Manager	Musawenkhosi Magongo	Coordinate stakeholder discussions, agree requirements, maintain the six-week plan, manage the repository and assemble the final submission.
Designer	Setsembele Magagula	Design the customer journey, USSD prompts, verification screen, status messages and staff dashboards; support usability evaluation.
Programmer	Bandile Lokothwayo	Implement USSD session handling, the backend, data model, mock services, duplicate protection, status checks, recovery and transaction history.
Tester	Sambulo Mdluli	Prepare test cases, verify balances and permissions, simulate interruptions, report defects and document evaluation results.

Table 6: Team roles and responsibilities

All team members will contribute to problem validation, design review, documentation and the final presentation.